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People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
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Final exam (2^h30)

Name :

Groupe:

Exercise 1. A household survey of gas consumption yielded the following results

Gas consumption	Frequency (f_i)	med-class (x_i)	Cum. Freq.	Rel.Cum. Freq.
0 – 9	1	4.5	1	0.01
10 – 19	2	14.5	3	0.03
20 – 29	1	24.5	4	0.04
30 – 39	5	34.5	9	0.09
40 – 49	8	44.5	17	0.17
50 – 59	16	54.5	33	0.33
60 – 69	19	64.5	52	0.52
70 – 79	20	74.5	72	0.72
80 – 89	17	84.5	89	0.89
90 – 99	11	94.5	100	1

(1)

- Complete the above statistical table.
- What is the average gas consumption of these household ? Calculate the variance, standard deviation and coefficient of variation.

Since the data are grouped, we have :

$$\mu = \bar{X} = \frac{\sum_{i=1}^7 f_i x_i}{\sum_{i=1}^7 f_i} = \frac{6650}{100} = \mathbf{66.5} \quad (0.25)$$

$$\sigma^2 = Var(X) = \frac{1}{\sum_{i=1}^N f_i - 1} \sum_{i=1}^N f_i (x_i - \mu)^2 = \frac{37800}{99} = \mathbf{381.82} \quad (0.5)$$

and

$$\sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i - 1} \sum_{i=1}^N f_i (x_i - \mu)^2} = \sqrt{381.82} = \mathbf{19.54} \quad (0.25)$$

The coefficient of variation is given by

$$CV = \frac{\sigma}{\mu} \times 100 = CV = \frac{19.54}{66.5} \times 100 = \mathbf{29.383\%} \quad (0.25)$$

- Give the Q_1 , Q_3 , D_5 and P_5 .

The first quartile Q_1 is located in the class 50 – 59 and the third quartile Q_3 is in 80 – 89 and both are given by

$$Q_1 = L_1 + \left(\frac{\frac{n}{4} - C f_{q_1}}{f_{q_1}} \right) \times w = 50 + \left(\frac{\frac{100}{4} - 17}{16} \right) \times 10 = \mathbf{55} \quad (0.5)$$

and

$$Q_3 = L_1 + \left(\frac{\frac{3n}{4} - Cf_{q_3}}{f_{q_3}} \right) \times w = 80 + \left(\frac{\frac{300}{4} - 72}{17} \right) \times 10 = \mathbf{81.765}. \quad (0.5)$$

Also, D_5 and P_5 are located in 60 – 69 and 30 – 39 respectively, and both are given by

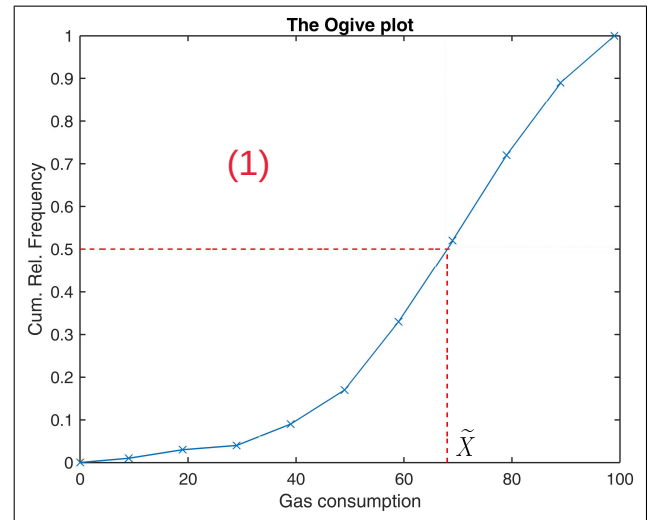
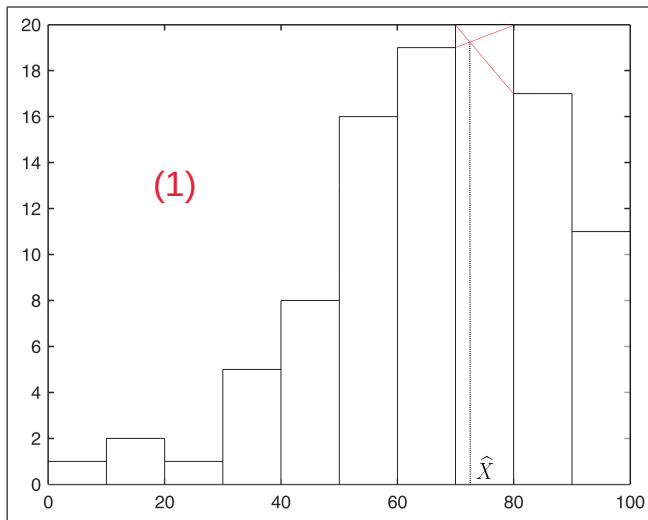
$$D_5 = L_1 + \left(\frac{\frac{5n}{10} - Cf_{d_5}}{f_{d_5}} \right) \times w = 60 + \left(\frac{\frac{500}{10} - 33}{19} \right) \times 10 = \mathbf{68.947} \quad (0.5)$$

and

$$P_5 = L_1 + \left(\frac{\frac{5n}{100} - Cf_{p_5}}{f_{p_5}} \right) \times w = 30 + \left(\frac{\frac{500}{100} - 4}{5} \right) \times 10 = \mathbf{32}. \quad (0.5)$$

Notice that $D_5 = Q_2 = \text{the Median } \tilde{X}$

4. Plot the histogram and the Ogive plot of the above distribution then determine the mode and median of the series graphically



5. Give the modale class then calculate the mode value.

Clearly, The highest frequency is for the class 70 – 79 to which the mode $\hat{X}$ belongs and it's given by

$$\hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w = 70 + \left(\frac{(20 - 19)}{(20 - 19) + (20 - 17)} \right) \times 10 = \mathbf{72.5} \quad (0.75)$$

Exercise 2. A statistician carries out a study on 200 executives of a company and seeks to study the link between the age of the executives and the monthly salary (in thousands of DA) that they receive. He presents his results in a contingency table, the statistical variable X representing age and the statistical variable Y monthly salary

$X \backslash Y$	[18, 22[	[22, 26[	[26, 30[	[30, 34[	[34, 38[	$n_{i.}$
[20, 28[	12	5	4	3	0	24
[28, 36[	7	11	9	4	1	32
[36, 44[	4	9	10	9	8	40
[44, 52[	1	6	12	11	15	45
[52, 60[	1	4	15	17	22	59
$n_{.j}$	25	35	50	44	46	200

1. Give the percentage of executives whose salary is greater than or equal to 26,000 DA.

$$\frac{n_{.3} + n_{.4} + n_{.5}}{N} \times 100\% = \frac{50 + 44 + 46}{200} \times 100\% = \mathbf{70\%} \quad (0.5)$$

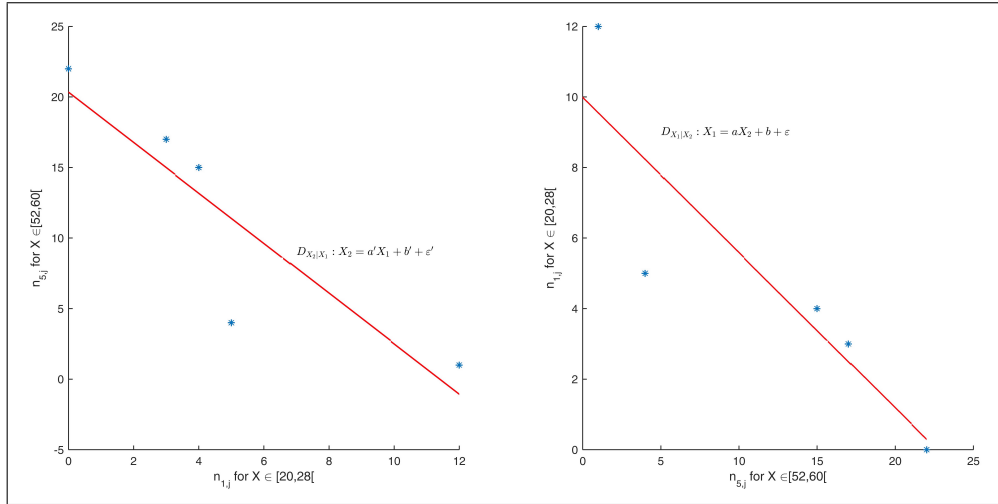
2. Among executives aged at least 36, give the percentage of those who receive less than 30,000 DA.

$$\frac{\sum_{i=3}^5 \sum_{j=1}^3 n_{ij}}{N} \times 100\% = \frac{4 + 9 + 10 + 1 + 6 + 12 + 1 + 4 + 15}{200} \times 100\% = \mathbf{31\%} \quad (0.5)$$

3. Give the conditional distribution of the age variable knowing that the salary received is in class [26,30[.

$X Y \in [26, 30[$	$[20, 28[$	$[28, 36[$	$[36, 44[$	$[44, 52[$	$[52, 60[$	(0.5)
<i>Frequency</i>	4	9	10	12	15	

4. Plot the scatter plot between age class [20, 28[and age class [52, 60[. Comment.



(1)

The scatter plot shows a relatively strong negative linear relation between the two classes (0.5)

5. Calculate the correlation coefficient and give the regression line between the two classes then plot it and give the value of the coefficient of determination.

$X_1 \equiv n_{1j} \text{ of } X \in [20, 28[$	12	5	4	3	0
$X_2 \equiv n_{52j} \text{ of } X \in [52, 60[$	1	4	15	17	22

The above scatter plot suggested the presence of linear correlation, hence, we choose pearson's coefficient of correlation to measure it.

We have

$$\sum x_{1i} = 24, \sum x_{2i} = 59, \sum (x_{1i} - \bar{X}_1)^2 = 78.8, \sum (x_{2i} - \bar{X}_2)^2 = 318.8, \sum (x_{1i} - \bar{X}_1)(x_{2i} - \bar{X}_2) = -140.2.$$

$$\begin{aligned} \rho(X_1, X_2) &= \rho(X_2, X_1) = \frac{Cov(X_1, X_2)}{\sigma_{X_1} \cdot \sigma_{X_2}} = \frac{\sum (x_{1i} - \bar{X}_1)(x_{2i} - \bar{X}_2)}{\sqrt{\sum (x_{1i} - \bar{X}_1)^2} \sqrt{\sum (x_{2i} - \bar{X}_2)^2}} \\ &= \frac{-140.2}{\sqrt{78.8} \times \sqrt{318.8}} = \mathbf{-0.88456} \end{aligned} \quad (1)$$

which indicates the existence of a strong negative linear correlation between the two variables. Set

$$X_1 = aX_2 + b + \varepsilon \quad \text{and} \quad X_2 = a'X_1 + b' + \varepsilon'$$

we have

$$\begin{aligned} a &= \frac{Cov(X_1, X_2)}{Var(X_2)} = \frac{\sum (x_{1i} - \bar{X}_1)(x_{2i} - \bar{X}_2)}{\sum (x_{2i} - \bar{X}_2)^2} = \frac{-140.2}{318.8} = \mathbf{-0.43977}, \\ b &= \bar{X}_1 - a\bar{X}_2 = \frac{24}{5} - (-0.43977) \times \frac{59}{5} = \mathbf{9.9893} \end{aligned}$$

and

$$a' = \frac{Cov(X_1, X_2)}{Var(X_1)} = \frac{-140.2}{78.8} = \mathbf{-1.7792},$$

$$b' = \overline{X_2} - a'\overline{X_1} = \frac{59}{5} - (-1.779) \times \frac{24}{5} = \mathbf{20.339}$$

thus

$$X_1 = -0.43977X_2 + 9.9893 + \varepsilon \quad \text{and} \quad X_2 = -1.7792X_1 + 20.339 + \varepsilon' \quad (1.5)$$

The coefficient of determination in both cases is

$$R^2 = \frac{V_E(X_1)}{Var(X_1)} = \frac{V_E(X_2)}{Var(X_2)} = \rho(X_1, X_2)^2 = \mathbf{0.78245} \quad (0.5)$$

Exercise 3. I— A diagnostic test for a disease is such that it (correctly) detects the disease in 90% of the individuals who actually have the disease. Also, if a person does not have the disease, the test will report that he or she does not have it with probability .9. Only 1% of the population has the disease in question. If a person is chosen at random from the population and the diagnostic test indicates that she has the disease

1. what is the conditional probability that she does, in fact, have the disease?

Set H : "a person has the disease" $\implies P(H) = 1\% = 0.01$, and E : "the test shows a positive result". Thus $P(E|H) = 90\% = 0.9$ and $P(E|H^c) = 1\% = 0.1$.

We want to find $P(H|E)$, using bayes' theorem, we have

$$P(H|E) = \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|H^c)P(H^c)} = \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.1 \times 0.99} = \mathbf{0.083333} \quad (1.5)$$

2. Are you surprised by the answer? Would you call this diagnostic test reliable?

surprisingly, the test does not appear to be reliable, because even with a positive result, the probability of having the disease is negligible

II— X and Y are independent, discrete random variables whose probability functions are given in the tables below:

x	1	2	3		y	1	2	3
$P(X=x)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$		$P(Y=y)$	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$

1. Compute the probability $P(X + Y = 4)$.

$$P(X + Y = 4) = P(X = 1, Y = 3) + P(X = 2, Y = 2) + P(X = 3, Y = 1)$$

$$\stackrel{X \text{ and } Y \text{ are indep.}}{=} P(X = 1)P(Y = 3) + P(X = 2)P(Y = 2) + P(X = 3)P(Y = 1)$$

$$= \frac{1}{3} \frac{1}{6} + \frac{1}{2} \frac{1}{6} + \frac{1}{6} \frac{1}{6} = \mathbf{0.25} \quad (1.5)$$

2. Compute the conditional probability $P(X \leq 2 | X + Y = 4)$

$$P(X \leq 2 | X + Y = 4) = \frac{P(X \leq 2, X + Y = 4)}{P(X + Y = 4)} = \frac{P(X = 1, Y = 3) + P(X = 2, Y = 2)}{P(X + Y = 4)}$$

$$= \frac{\frac{1}{3} \frac{1}{6} + \frac{1}{2} \frac{1}{6}}{0.25} = \mathbf{0.556} \quad (1)$$

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3. Give the expectation of $g(X, Y) = X^2Y$.

Since X and Y are independent, we have $E(g(X, Y)) = E(X^2Y) = E(X^2)E(Y)$ with

$$\begin{aligned}E(X^2) &= 1^2 \times \frac{1}{3} + 2^2 \times \frac{1}{2} + 3^2 \times \frac{1}{6} = \mathbf{3.8333} \\E(Y) &= 1 \times \frac{2}{3} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} = \mathbf{1.5}\end{aligned}\tag{1.5}$$

Therefore, $E(g(X, Y)) = \mathbf{5.75}$

4. Give the CDF of the $\min(X, Y)$

Set $Z = \min(X, Y)$, then, the CDF of Z is given by

$$\begin{aligned}F_Z(z) &= P(Z \leq z) = 1 - P(Z > z) \\&= 1 - P(\min(X, Y) > z) \\&= 1 - P(X > z \text{ and } Y > z) \\&= 1 - P(X > z)P(Y > z) \\&= 1 - (1 - P(X \leq z))(1 - P(Y \leq z)) \\&= 1 - (1 - F_X(z))(1 - F_Y(z))\end{aligned}\tag{1.5}$$